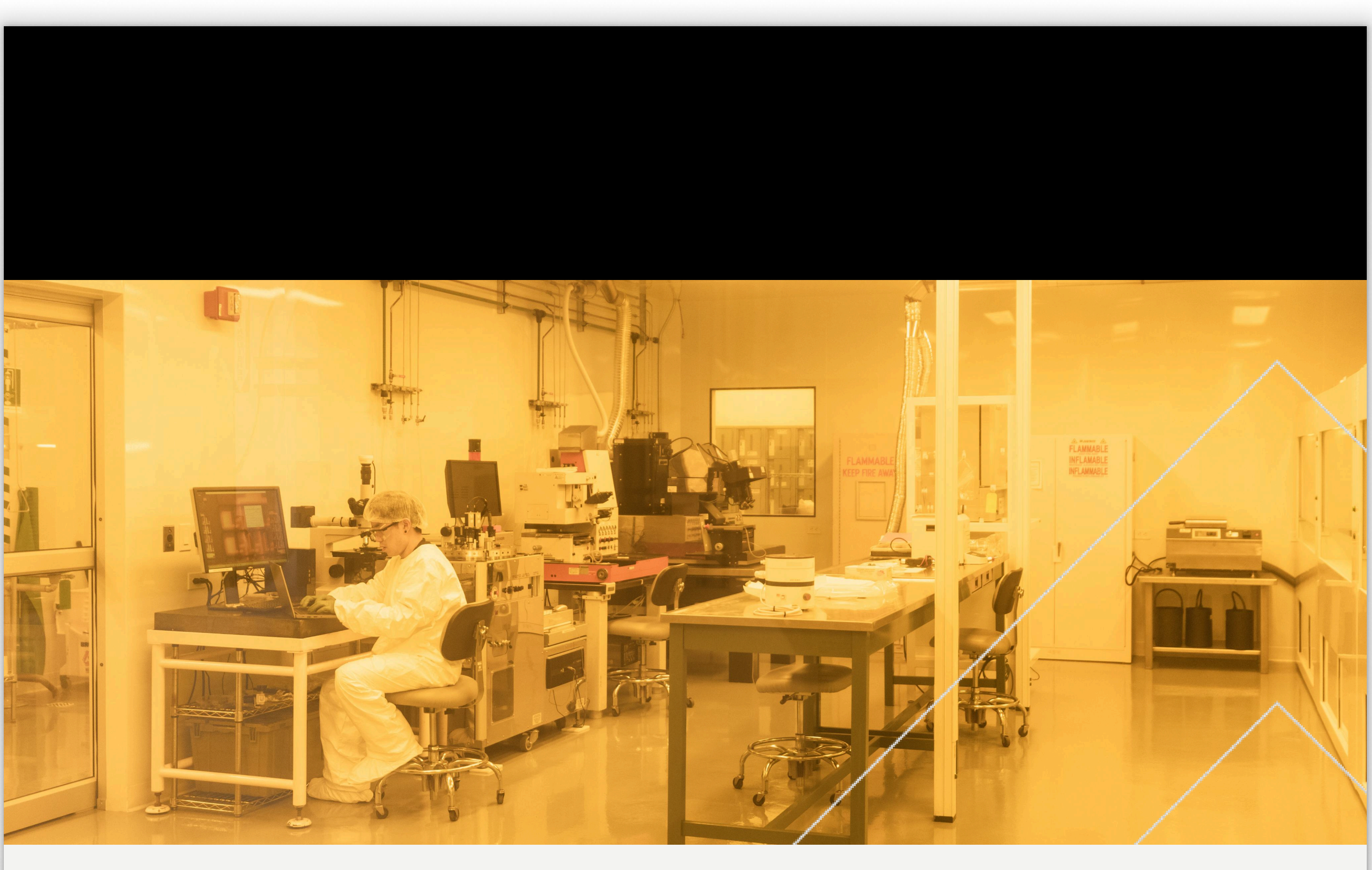

The Toronto Nanofabrication Centre (TNFC) is an interdisciplinary research and service centre at the University of Toronto.

The centre offers global research leadership by providing access to state-of-the-art nanofabrication facilities, collaborative research networks, advanced educational opportunities and information exchange events for registered users and clients.

Key Research Areas

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Microelectronics **Microfluidics** **MEMS/NEMS**

Photonics **Plasmonics** **CMOS Processing**

Photovoltaics **Integrated Optics** **Nanomaterial Processing**

Facilities

Pratt Microfabrication Facility

The Pratt Microfabrication Facility supports a wide range of silicon-based fabrication activities including photolithography, oxidation, thin film deposition, plasma etching, wet etching, device processing, packaging and analysis.

The 400m2 facility is located on the 4th floor of the Pratt engineering building at the University of Toronto, and includes both a class 10,000 (ISO Class 7) microfabrication processing area and a class 1,000 (ISO Class 6) photolithography processing area.

Wallberg Nanolithography Facility

The Wallberg Nanolithography Facility (an investment of over $6 million) houses the state-of-the-art Raith EBPG5000+ Electron Beam Lithography System. The 108 m2 facility is located in the basement of the Wallberg building at the University of Toronto in order to achieve very low mechanical vibration of <10 μm/s.

To ensure exceptional performance, the system is enclosed in an environmental chamber maintaining temperature stability of 21 ± 0.25 °C, 0.1 °C/hr. The 58 m2 Class 100 cleanroom lab space consists of the environmental chamber, control room, processing lab and gowning room, and contains a complete suite of equipment to perform nanolithography from CAD design to exposed resist development and analysis.

Capabilities

Electron Beam Lithography
Etching +
Photolithography +
Dicing & Packaging +
Deposition +
Imaging & Characterization

Services

User Access

All users must complete an ECE User Access Form and attend mandatory training sessions before using TNFC facilities.

All new and existing non-ECE staff members and students will be required to additionally fill out an Attestation Form. Both forms are to be completed and sent to tnfc@utoronto.ca. Researchers will be charged for the use of the facilities on a per-minute basis as logged by our LMACS Access System, and invoices will be distributed on a quarterly basis.

For non-UofT academic and industrial researchers, please send an email to tnfc@utoronto.ca to work out an arrangement to come onsite for a short-term visit.

Contact TNFC **Learn More**

Client Services

TNFC offers a wide range of fabrication and characterization services to academic and industrial clients. Many researchers require infrequent access to TNFC capabilities in order to prototype or analyze a small number of devices. For these cases, user access and training may not be practical. TNFC technical staff offer experienced service to perform fabrication and characterization processes on behalf of the user for a predetermined fee.

TNFC is also a proud member of the CMC FACT Network, which leverages experts and state-of-the-art toolsets from micro/nano fabrication laboratories at Canadian universities for custom fabrication and characterization services.

To request a service or for more information:

Contact TNFC

Our Partners

Granting Agencies

Partner Organizations

Canadian Nanofabrication Facilities

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